

Exp no:6

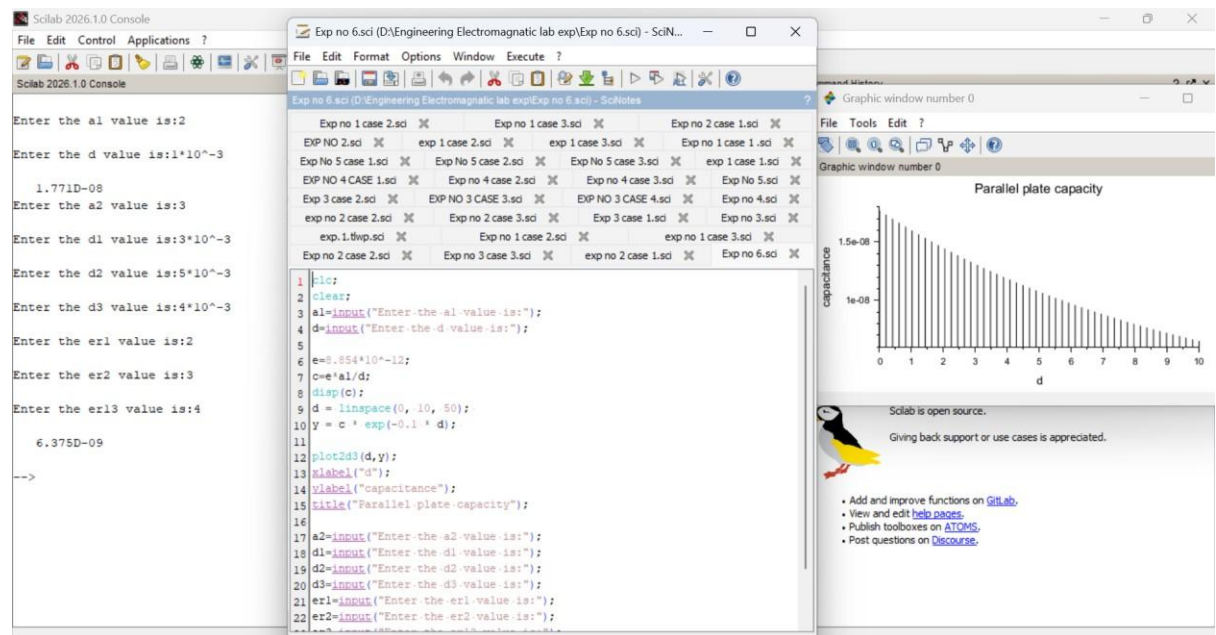
Determination and plotting of Capacitance of parallel plate capacitor in three different dielectric media with specific areas, dielectric thickness and various dielectric constant.

Aim:

To write program for capacitance of parallel plate capacitor in three different dielectric media using SCILAB.

Software required:

- SCILAB version 6.0.1



Exp No: 06

$$d_1 = 2 \text{ cm}^2$$

$$d_2 = 3 \text{ cm}^2$$

$$d_3 = 1 \text{ mm}$$

$$d_4 = 5 \text{ mm}$$

$$d_5 = 4 \text{ mm}$$

$$\epsilon_{r1} = 2$$

$$\epsilon_{r2} = 3$$

$$\epsilon_{r3} = 4$$

$$C = \frac{\epsilon_r \epsilon_0 A}{d}$$

$$\epsilon_r = 1 \text{ and } 1.5$$

Capacitance of Air

$$C = \frac{\epsilon_0 A}{d}$$
$$= \frac{8.854 \times 10^{-12} \times 2}{1 \times 10^{-3}}$$

$$= 1.7708 \times 10^{-8}$$

$$C = 1.771 \times 10^{-8}$$

Capacitance of three dielectric layer

O/p

$$C = \frac{\epsilon_0 A}{\left(\frac{d_1}{\epsilon_{r1}}\right) + \left(\frac{d_2}{\epsilon_{r2}}\right) + \left(\frac{d_3}{\epsilon_{r3}}\right)}$$

$$= \frac{d_1}{\epsilon_{r1}} = \frac{3 \times 10^{-3}}{2} = 1.5 \times 10^{-3}$$

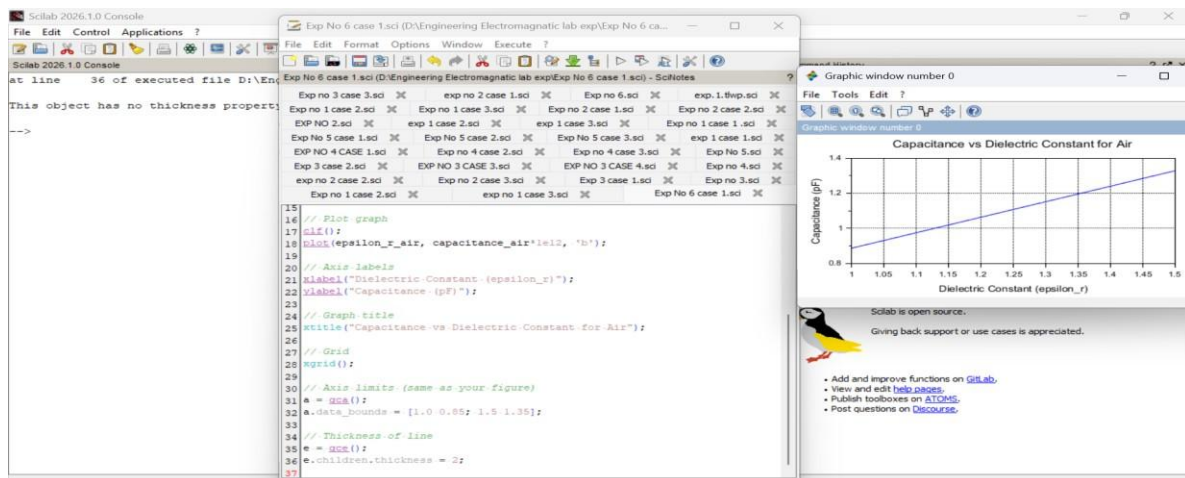
$$= \frac{d_2}{\epsilon_{r2}} = \frac{5 \times 10^{-3}}{3} = 1.667 \times 10^{-3}$$

$$= \frac{d_3}{\epsilon_{r3}} = \frac{4 \times 10^{-3}}{4} = 1 \times 10^{-3}$$

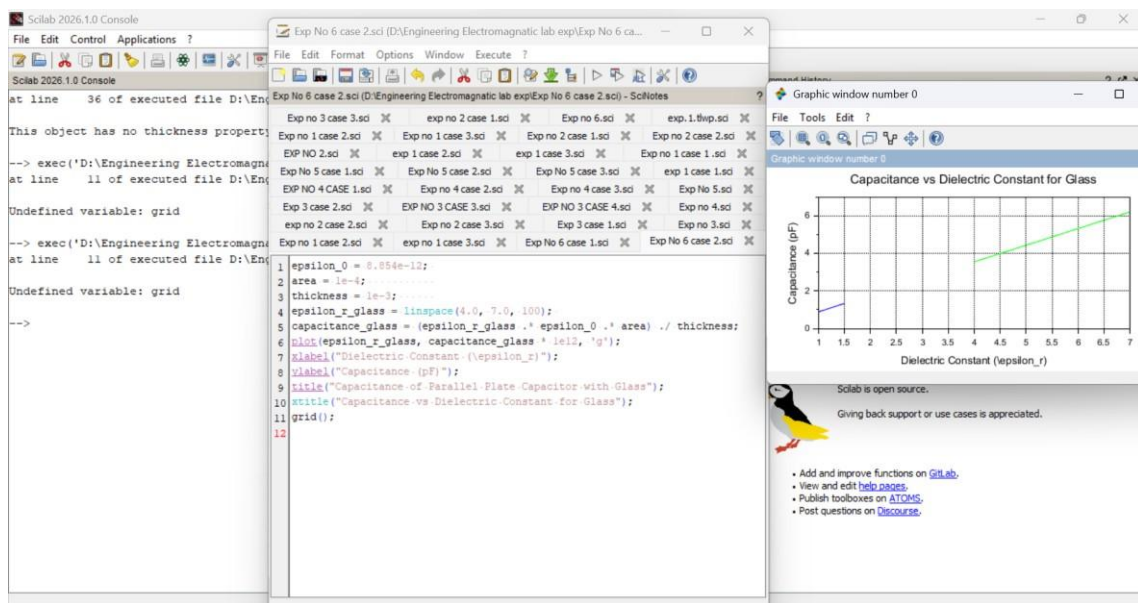
$$C = 6.375 \times 10^{-9}$$

Determination and plotting of Capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constant

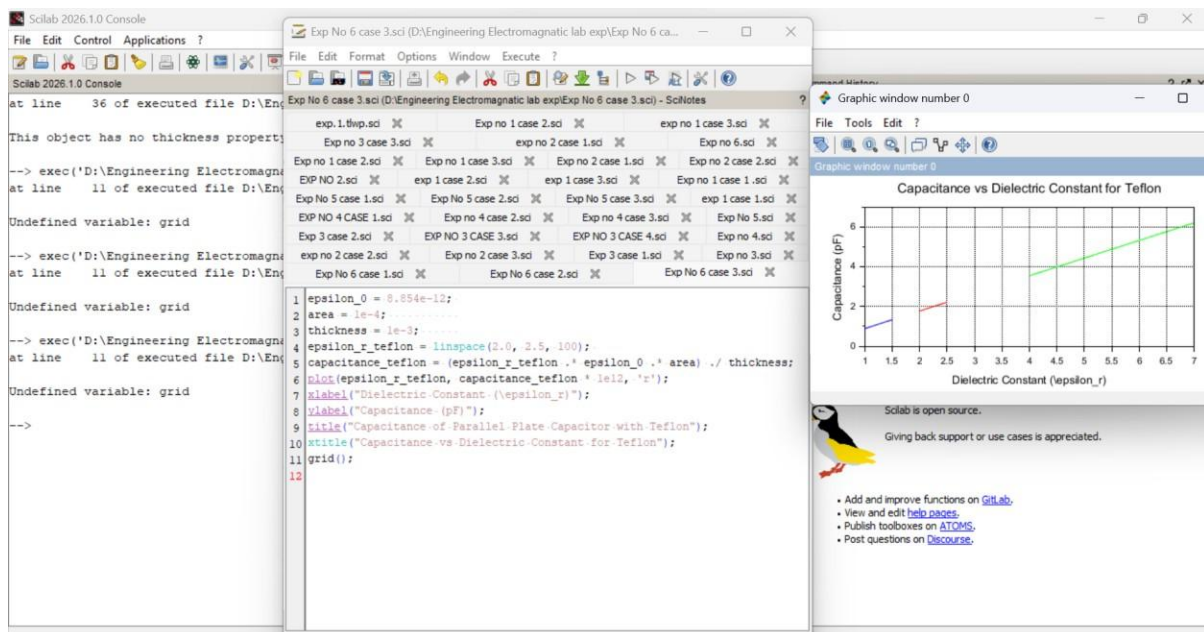
CODE:



Case 2. Glass:



Case 3. Teflon:



Result :

Thus the program was verified for capacitance of parallel plate capacitor in three different dielectric media using SCILAB was successfully.